



IPS ACADEMY INDORE

**INSTITUTE OF ENGINEERING & SCIENCE
DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY**

SESSION JUL-DEC 2025

MID TERM TEST I

BTECH III YEAR V SEM

TIME: 1 HR 15 MIN

IFC-CB-001 Block Chain Technology

MAX MARKS: 20

Q. No.	Question Description	CO	BL	PO	Marks Allotted
Q.1(A)	What is ECC and why it is used in Blockchain?	1	1	1,2,3	5
(B)	Define Blockchain. List and explain any 4 applications of blockchain.	1	2	1,2,3	5
OR					
Q.1(A)	Explain the working of ECDSA in blockchain encryption and decryption.	1	2	1,2,3	5
(B)	What is a hashing algorithm? Explain SHA-256 with example.	1	4	1,2,3	5
Q.2(A)	Define Blockchain mining? Why it is essential for blockchain networks?	2	1	1,2,3	5
(B)	What is Consensus Mechanism? Explain its importance in blockchain.	2	2	1,2,3	5
OR					
Q.2(A)	Explain Byzantine General Problem with a suitable example.	2	2	1,2,3	5
(B)	Differentiate between Proof-of-Work and Proof-of-Stake algorithms	2	4	1,2,3	5

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SESSION JULY-DEC 2025
MID TERM TEST I

BRANCH- CSIT

CLASS- BTECH III YEAR V SEM

PCC-CI07

Database Management System

TIME: 1 HR 15 MIN

MAX MARKS: 20

Q. No.	Question Description	CO	BL	PO	Marks Allotted
Q.1 (A)	What is Database System Architecture? Explain the different levels of architecture with neat diagram.	1	2	1,2	5
(B)	Draw an E-R diagram for an Online Examination System including weak entities and aggregation.	1	3	1,3	5
	OR				
Q.1 (A)	Explain the advantages of using a Database Management System (DBMS) over a traditional file system.	1	2	1,2	5
(B)	With a suitable example, explain how generalization and specialization concepts are applied in E-R modeling to handle redundancy and improve schema efficiency.	1	2	1,2,3	5
Q.2 (A)	Define Candidate Key, Primary Key, Foreign Key and Alternate Key with suitable examples. How does Primary Key ensure uniqueness and integrity?	2	2	1	5
(B)	Explain tuple-oriented relational calculus and domain-oriented relational calculus with their syntax and example.	2	3	1	5
	OR				
Q.2 (A)	Write relational algebra queries using given schemas: Schema: Student(rollno, name, course_id) Course(course_id, course_name) Queries: a) Retrieve the names of all students. b) List all courses offered in the database. c) Find the names of students along with the courses they are enrolled in. d) Retrieve roll numbers of students who are enrolled in the course 'DBMS'. e) List all student names who are enrolled in any course.	2	3	2,3	5
(B)	How do integrity constraints ensure correctness of data in a relational database?	2	2	1	5



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SESSION JUL-DEC 2025

MID TERM TEST I

BTECH III YEAR V SEM

PCC-CI09 Formal Language and Automata

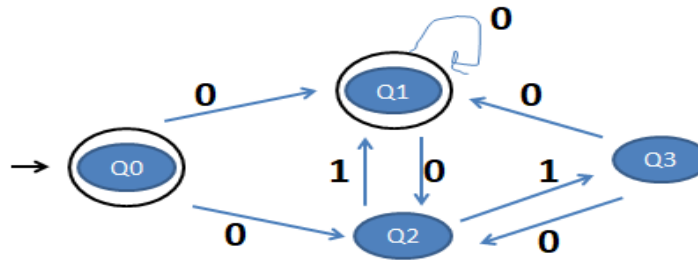
TIME: 1 HR 15 MIN

MAX MARKS: 20

Q. No.	Question Description	C O	B L	P O	Marks Allotted
Q.1 (A)	Explain Deterministic Finite Automata (DFA) with the help of Example.	1	2	2	5
(B)	Design NFA which accepts set of all strings over $\{a, b\}$, which starts with 'ab' and ends with 'bb'	1	3	2,3	5

OR

Q.1 (A)	Discuss Moore Machine with the help of Example.	1	2	2	5
(B)	How would you convert the given NFA to DFA?				



Q.2 (A)	Explain Union, Intersection and Concatenation closure properties of regular expression.	2	2	2	5
(B)	Construct the Regular Expression for the language accepting the strings which are starting with '11' and ending with '01', over the set $\Sigma = \{0, 1\}$.	2	3	2,3	5

OR

Q.2 (A)	Explain Pumping Lemma for Regular Language.	2	2	2	5
(B)	Construct the Regular Expression for the language accepting the strings, such that the 3 rd character from right end of the string is always 'a'. $\Sigma = \{a, b\}$.	2	3	2,3	5



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SESSION JUL-DEC 2025

MID TERM TEST I

BTECH II YEAR III SEM

TIME: 1 HR 15 MIN

PCC-CI08 OPERATING SYSTEM

MAX MARKS: 20

Q. No.	Question Description	CO	BL	PO	Marks Allotted
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Q.1(A) What is a Process Control Block (PCB)? List its main components and explain their role during context switching. 1 1 1 5

(B) *SRTF – Preemptive SJF* 1 3 1,2 5

Apply Shortest Remaining Time First (SRTF) and calculate the Average Waiting Time of the system.

Process	AT	BT
P1	0	7
P2	1	4
P3	2	3
P4	3	2

OR

Q.1(A) Draw and explain the Process State Transition Diagram. Mention when a process moves to each state. 1 3 1 5

(B) *Round Robin – $TQ = 3$* 1 3 1,2 5

Using Round Robin (Time Quantum = 3 units), calculate the Average Waiting Time of the system.

Process	AT	BT
P1	0	5
P2	1	7
P3	2	3
P4	4	6

Q.2(A) Describe Peterson's Solution for two processes. Show how it satisfies Mutual Exclusion, Progress, and Bounded Waiting. 2 2 1 5

(B) Consider a system with 3 processes (P1, P2, P3) and 3 resource types (A, B, C). The following Allocation and Need matrices are given. 2 3 1,2 5

Allocation Matrix:

Process	A	B	C
P1	0	1	0
P2	2	0	0
P3	3	0	3

Need Matrix:

Process	A	B	C
P1	7	4	3
P2	1	2	2
P3	6	0	0

Available = (3, 3, 2)

- Construct the **Resource Allocation Graph (RAG)** for the above system.
- Using the RAG, determine whether the system is in a **safe state or not**. Explain briefly.

OR

Q.2(A) What is a Semaphore? Differentiate between Counting Semaphore and Binary Semaphore with suitable examples. 2 4 1 5

(B) Given the following system snapshot: 2 3 1,2 5

Process	Allocation (A, B, C)	Max (A, B, C)
P0	(0, 1, 0)	(7, 5, 3)
P1	(2, 0, 0)	(3, 2, 2)
P2	(3, 0, 2)	(9, 0, 2)
P3	(2, 1, 1)	(2, 2, 2)

Available Resources: (3, 3, 2)

Calculate:

- The Need Matrix (Max - Allocation)
- Is the system currently in a safe state? If yes, find the safe sequence.



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SESSION JUL-DEC 2025

MID TERM TEST I

BTECH III YEAR VSEM

TIME: 1 HR 15 MIN

PCC-CI10

Software Engineering & Project Management

MAX MARKS: 20

Q. No.	Question Description	CO	BL	PO	Marks Allotted
Q.1(A)	Explain Waterfall Model and Spiral Model for software development.	1	2	1	5
(B)	Describe important characteristics of software process. Differentiate between a software product and a software process. OR	1	2	1	5
Q.1(A)	Explain phases of the RAD Model. Discuss situations where the RAD Model is useful.	1	2	1	5
(B)	Explain the Scrum framework in detail. Mention its roles, artifacts, and ceremonies.	1	2	1	5
Q.2(A)	Compute the function point value for a project with the following information domain characteristics: Number of user inputs: 32 Number of user outputs: 60 Number of user inquiries: 24 Number of files: 8 Number of external interfaces: 2 Use the following average complexity weight factors: Input = 4, Output = 5, Inquiry = 4, File = 10, External Interface = 7 Assume that all complexity adjustment values are average.	2	3	2	5
(B)	Explain Lines of Code (LOC)? Differentiate between LOC and FP estimation techniques. OR	2	2	2	5
Q.2(A)	A software project is estimated to have 32 KLOC . Use Basic COCOMO (Organic Mode) to calculate: 1. Effort (in Person-Months) 2. Development Time (in Months) 3. Average Staff Size Constants: $a=2.4$, $b=1.05$, $c=2.5$, $d=0.38$	2	3	2	5
(B)	Define software reliability. Explain MTTF, MTTR, and MTBF with formulas.	2	2	2	5

IPS ACEDEMY
INSTITUTE OF ENGINEERING & SCIENCE
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
BRANCH- CYBER,MECHANICAL, CSIT, IoT
SESSION JULY-DEC 2025
MID TERM TEST I

CLASS- B.TECH. III YEAR V SEM

Japanese Language- HSMC HS 05

TIME: 1 HR 15 MIN

MAX MARKS: 20

Q. No.	Question Description	C O	B L	PO	Marks Allotted
Q.1 (A)	Identify the numbers. a) san jyū ni b) roppyaku go jyū hachi c) kyū sen kyū hyaku kyū jyū kyū d) hassen shichi jyū ichi e) go hyaku shi jyū roku	1	2	1,2, 3	5
(B)	Write any five greetings in Japanese (using Hiragana script) and write their meanings in English.	1	1	1,2, 3	5
	OR				
Q.1 (A)	Name the following:- a) The script used in Japanese Language for writing the words borrowed from other foreign languages. b) The pictorial script in Japanese Language. c) The script used for writing original Japanese Language words. d) The name by which Japanese Language is known by. e) The total number of alphabets in the script which we have learned.	1	2	1,2, 3	5
(B)	Mention any 5 things/ persons (in Hiragana) which you can spot in the picture given below.	1	1	1,2, 3	5

					
Q.2(A))	Write the names of days of a week in Japanese.	2	1	1,2, 3	5
(B)	Write the first ten days of a month in Japanese.	2	1	1,2, 3	5
	OR				
Q.2(A))	Write time in Japanese. (any five) a). 10:30 P.M b). 12:02 A.M c). 4: 23 P.M d). 6:54 P.M e). 8:42 A.M f). 7:00 P.M	2	1	1,2, 3	5
(B)	Write the following words in Hiragana along with their meanings. (any five) a). dōbutsu b). bijyutsukan c). kissaten d). kōen e). shyashin f). gakkō	2	1	1,2, 3	5

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IPS ACADEMY
INSTITUTE OF ENGINEERING & SCIENCE
MID TERM TEST I
SESSION JULY-DEC 2025
ALL BRANCHES

CLASS- BTECH III Year/ VSem

HSMC-HS05 (French)

TIME: 1 HR 15 MIN MAX MARKS: 20

Q. No.	Question Description	CO	BL	PO	Marks
		1	3	9,10	5

Q.1 (A) Choisissez la bonne réponse:

(i) Comment t'appelles-tu?

- a. Je vais bien, merci.
- b. J'habite à Indore.
- c. Je m'appelle Sakshi.

(ii) Mettez les jours en ordre:

- a. Lundi, mardi, mercredi, jeudi, vendredi, samedi, dimanche
- b. Lundi, mardi, jeudi, vendredi, mercredi, samedi, dimanche
- c. Lundi, mardi, mercredi, jeudi, dimanche, samedi, vendredi

(iii) J'habiteParis.....France.

- a. en / au
- b. à / en
- c. en / à

(iv) Dans ma trousse il y a.(1).....(2)..... et (3).....(**choose 3**)

- a. une chaise
- b. un crayon
- c. une classe
- d. un stylo
- e. une gomme
- f. un TV

(v) Ecrivez la forme féminine des nationalités:

- a. Indien (m)(f)
- b. Français (m)(f)

(B) Conjuguez le verbe <<Être et Avoir>> avec la traduction. 1 3 9,10 5

OU (OR)

Conjuguez le verbe <<Avoir et Aller>> avec la traduction.

Q.2(A)Écrivez les nombres en français. 1 5,6 9,10 5

11 - 9 - 7 - 34 - 20 - 45 - 50 - 30 - 18 - 5

(B)Traduisez en anglais: 1 5,6 9,10 5

- 1. Bonsoir!
- 2. Je parle anglais.
- 3. Elle est américaine.
- 4. Nous sommes canadiens.
- 5. Aujourd'hui
- 6. Nous avons deux crayons rouge.
- 7. Il est espagnol.
- 8. Je vais au café.
- 9. Ils vont au centre commercial.
- 10. Mon stylo est noir.

OU (OR)

(B) Réécrivez les phrases en bon ordre:

- 1. chocolat / j' / les/ aime /.
- 2. rouges /il / fruits/ adore/ les /.
- 3. va/ centre/elle/ au / commercial.
- 4. regardent / le /elles/ film/.
- 5. grande/maison/ ma/et/ belle/est/.

****<<Je vous souhaite Bonne Chance>>****

<<Je vous souhaite Bonne Chance>>

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IPS ACADEMY
 Institute of Engineering and Science, Indore (M.P.)
 (A UGC Autonomous Institute, affiliated to RGPV Bhopal)
 Department of Computer Science & Engineering
B.Tech , III Yr. V Sem, Mid Test Examination- I, Sept. -2025
Soft Skills & Interpersonal Communication [HS05]

Max marks: 20

Time: 60 min.

S. No.	Question Description	Marks Allocated	CO	BL	PO
1	A. Write the importance of soft skills and interpersonal communication skills in the field of computer science and Engineering. B. Write short notes on informal (Grapevine) communication in an organization.	5 5	1	1,2	1,10
	OR A. “Silence speaks louder than words” explain the statement in the light of non-verbal communication. B. Write the difference between verbal and non-verbal communication.	5 5	1	1,2	1,10
2	A. Listening is more important than hearing in professional life. Elucidate the statement. B. Write short notes on culture as communication.	5 5	2	1,2	1,6,10
	OR A. Describe the types of listening in detail. B. Differentiate between hearing and listening.	5 5	2	2,5	1,6,10